The second law

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Explain the direction of heat transfer and the limits of energy conversion.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 5.*
No. Equilibrium must lie between 13°C and 55°C under the assumptions. Check readings or an unmodeled input.

02. Worked example: explain the model
Net heat flows hot to cold; equilibrium has equal temperatures under suitable isolation.

03. Read and interpret the model
Hot to cold; temperatures move toward a common value if isolated.

04. Apply the physics
25 J by energy conservation.

05. Explain your reasoning
A cyclic heat engine cannot convert all heat from a single reservoir to work; some heat rejection/entropy accounting is required.

06. Review: connect an earlier idea *Review: Unit 4, Day 18.*
4 J remains unaccounted. Seek another storage/transfer or check measurements.

07. Exit check
No; refrigerator is not passive/spontaneous cold-to-hot transfer; work enables it.

08. STRETCH: test the model
The passive pack absorbs energy from a warmer object. The refrigerator uses work to transfer energy from its colder compartment to the warmer room; rejected energy includes both the removed heat and the work input.

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01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 5.*
Water gains 21000 J; metal c≈1167 J/(kg·°C). Equal energy loss/gain applies under the assumptions.

02. Worked example: explain the model
Net heat flows hot to cold; equilibrium has equal temperatures under suitable isolation.

03. Read and interpret the model
Hot to cold; temperatures move toward a common value if isolated.

04. Apply the physics
250 J; COP 4. COP is not a heat-engine efficiency and may exceed 1.

05. Explain your reasoning
A cyclic heat engine cannot convert all heat from a single reservoir to work; some heat rejection/entropy accounting is required.

06. Review: connect an earlier idea *Review: Unit 4, Day 18.*
Energy=7.2 J; average opposing force≈9 N.

07. Exit check
Energy conserved; temperature-difference opportunity for work is reduced/eliminated at equilibrium.

08. STRETCH: test the model
280 J rejected; includes both removed heat and input work.